Ripple

From a news headline to a bank's loss range, with the evidence attached.

Reads news and social text, scores each event, checks the score against past market moves, and traces the event through a bank's portfolio.

Mayank Rathore
VIT Bhopal

Case study

AI/NLP risk engine with a stress-test module and a rebalancer module

S&P Global & Crisil Code to Connect

A sentiment score is not a loss number, and four gaps explain why

Why plain scores fall short

Not tied to prices

Impact scores are usually hand-set. Nobody checks them against what markets did after similar events.

Shocks are guessed

Stress tests apply one round number, like a 10% equity drop, to every kind of event.

Exposure is one hop

Losses travel through suppliers, sectors and counterparties the headline never names.

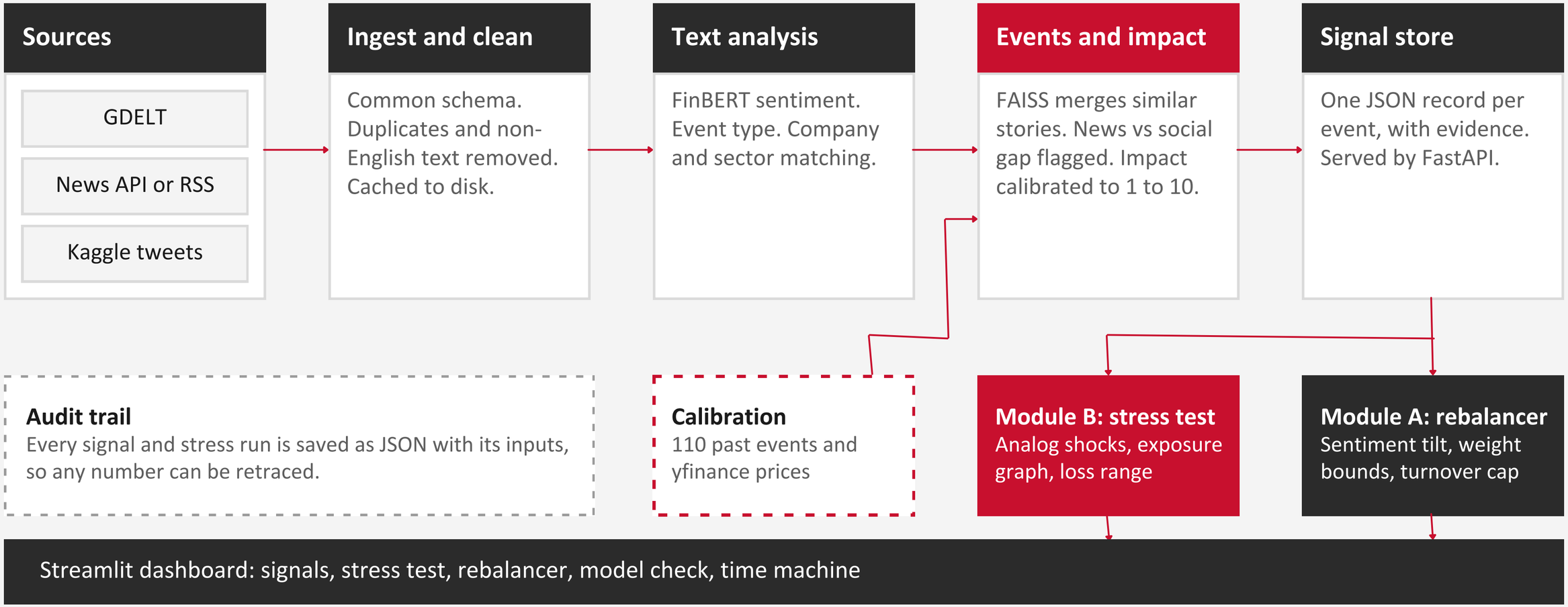
Repeats look like signal

Fifty articles on one story count as fifty events, and news and social text can disagree.

What Ripple does

1. Reads GDELT, news feeds and tweets; merges repeat stories into one event.
2. Scores sentiment, event type and impact; flags news vs social gaps.
3. Checks impact against price moves after 110 past events.
4. Takes shocks from similar past events and spreads them through an exposure graph.
5. Revalues a bank book of loans, bonds and derivatives, and shows a loss range. The same signals tilt an index.

One pipeline turns text into a signal that two modules can use



Signal fields `sentiment_score, event_type, impact_score, confidence, divergence_flag, evidence`

CPU only. Cache fallback. Offline replay.

Four choices keep the numbers honest, and the score is tested against history

Impact checked against history

Isotonic regression maps the raw score to the price move seen after 110 past events (77 train, 33 test). It can only rise with the score.

The score has to earn its meaning

Shocks from similar events

FAISS finds analogs such as 2008, 2020 and 2023. With fewer than 3, the shock is pulled toward a stated prior.

A range, not one guess

Contagion through a graph

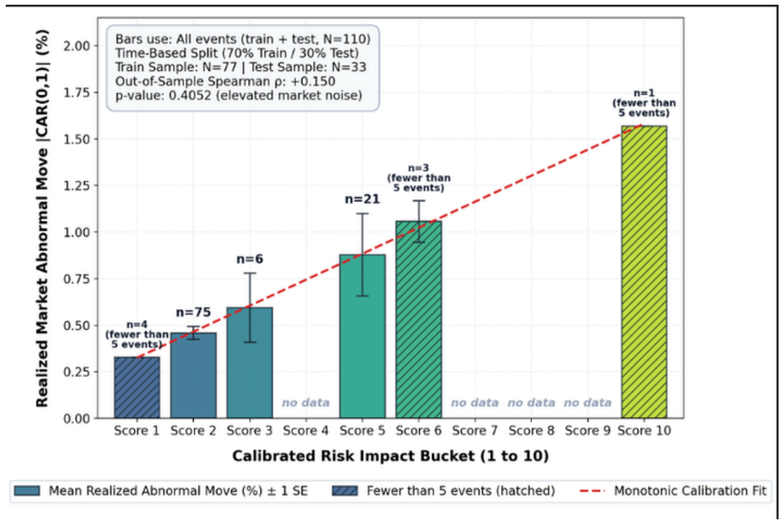
Companies, sectors and counterparties are linked. The shock halves at each hop, up to 2 hops.

Second-order exposure shows up

Rebalancer with limits

Tilt = sentiment x impact/10 x confidence, fading over time. Weight bounds and a turnover cap apply.

One headline cannot swing the index



What the chart shows

- Scores 2, 3 and 5 have enough events, and the mean move rises: about 0.46%, 0.60%, 0.88%
- Out-of-sample Spearman +0.15, p = 0.41 (N = 33)
- Scores 1, 6 and 10 have under 5 events; 4, 7, 8, 9 have none

Stack: Python 3.11 | FinBERT | sentence-transformers | FAISS | NetworkX | yfinance | scikit-learn | FastAPI | Streamlit | Plotly

One event, one bank book: a median loss of $159.3M, with a range of $146M to $172M

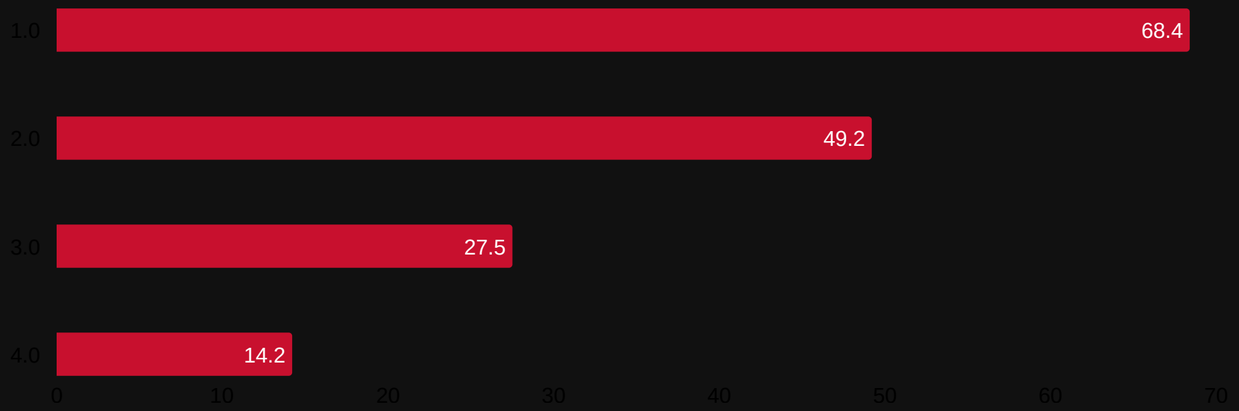
$159.3M

median loss, 1.87% of an $8.54B book (42 positions)

Range across shock estimates



Median loss by asset class, USD m



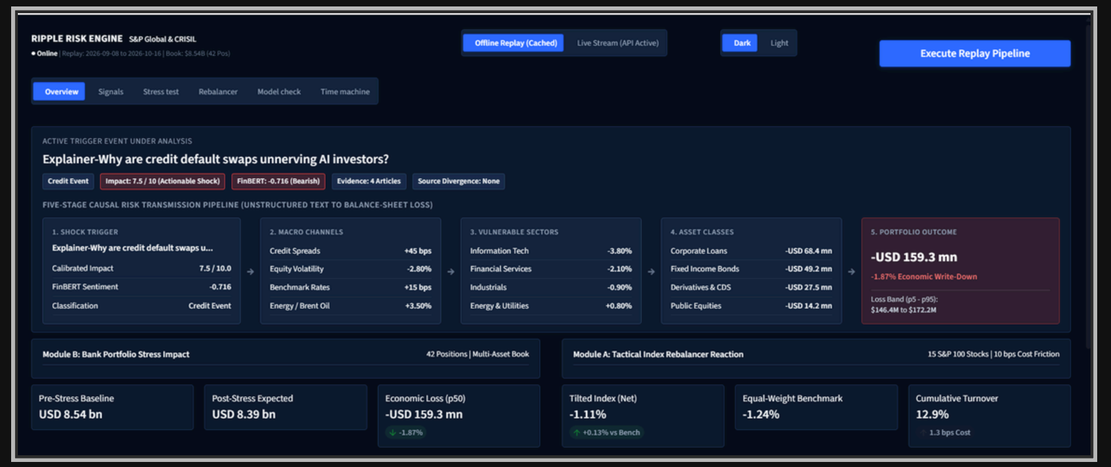
Source: dashboard Overview page. Replay 2026-09-08 to 2026-10-16, offline cache. Synthetic book.

What drove it

- Event: credit event, impact 7.5 of 10, FinBERT -0.716, 4 articles
- Macro moves: credit spreads +45 bps, rates +15 bps, oil +3.5%
- Hardest hit sectors: information technology -3.8%, financial services -2.1%

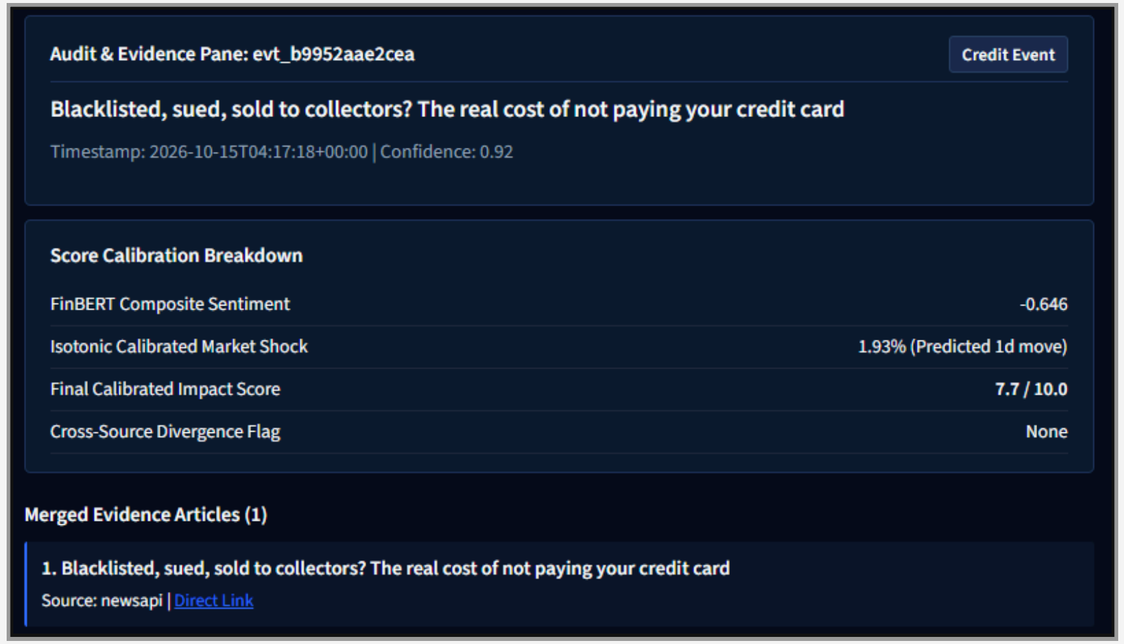
Module A: index rebalancer

Tilted index -1.11% vs equal-weight -1.24%, a gap of +0.13 points. Turnover 12.9%, cost 1.3 bps, 15 S&P 100 stocks. One window, one event, small gap.



A risk team gets a traceable loss range per event instead of a sentiment number

Ten articles on the same story	→	**One event, with every source listed**
A sentiment number	→	**An impact score checked against past price moves**
One round-number shock	→	**Shocks from similar past events, shown as a range**
Exposure to the named company only	→	**Second-order exposure through sector and counterparty links**
A result nobody can retrace	→	**Every run saved as JSON with its inputs**
Index weights set by hand	→	**Sentiment tilt with weight bounds and a turnover cap**



What the evidence pane shows

- Score breakdown: FinBERT sentiment, isotonic market shock, final impact
- The source article, with a direct link
- Filter by impact: 12 of 143 signals score 7.5 or higher

Who uses it: credit risk teams (loss range per event), treasury (rate, FX and commodity shocks), index managers (sentiment tilt with limits).

The method is shown end to end; the numbers need more data before anyone relies on them

Where it is thin

- Calibration is weak: 110 events, out-of-sample correlation +0.15 (p = 0.41)
- Bank book is synthetic
- Bonds repriced by duration only, no convexity
- Balance sheet held still, no hedging
- Social text is a public Kaggle set, not live

Next

- Add many more events and re-test the score
- Add convexity, then option gamma and vega
- Read company filings as another text source
- Backtest the rebalancer over a longer period, with trading costs

Code

github

Contact

mayank.23bai10291@vitbhopal.ac.in

Demo video

Linked in the README